Part B Team Integration

Consolidated Class Inventory

ID	Class	Purpose	Related use cases	Contributor
C01	User	Common account information and authentication operations	UC22	Omar Abdalla, Alan Dsouza, Omar Alkhatib
C02	Customer	Customer profile, prescriptions and purchasing	UC01, UC06, UC13, UC14, UC15, UC21	Omar Abdalla, Alan Dsouza
C03	Pharmacist	Prescription review and fulfillment	UC11, UC12, UC24	Omar Abdalla, Alan Dsouza
C04	Prescription	Uploaded prescription information and review status	UC01, UC06, UC11, UC12, UC24	Omar Abdalla
C05	Order	Purchase information, payment eligibility and status	UC03, UC13, UC14, UC15	Omar Abdalla, Alan Dsouza, Omar Alkhatib
C06	Medication	Medication identity, pricing and prescription requirement	UC04, UC05, UC08, UC16, UC17	Alan Dsouza, Omar Alkhatib
C07	PharmacyAdmin	Maintenance of medication and stock records	UC04, UC05, UC08	Omar Alkhatib
C08	Inventory	Collection of stock items and availability checking	UC04, UC05, UC08	Omar Alkhatib
C09	StockItem	Available quantity and reorder information	UC04, UC05, UC08	Omar Alkhatib
C10	OrderItem	Requested medication, quantity and price in an order	UC03, UC15	Proposed team integration addition

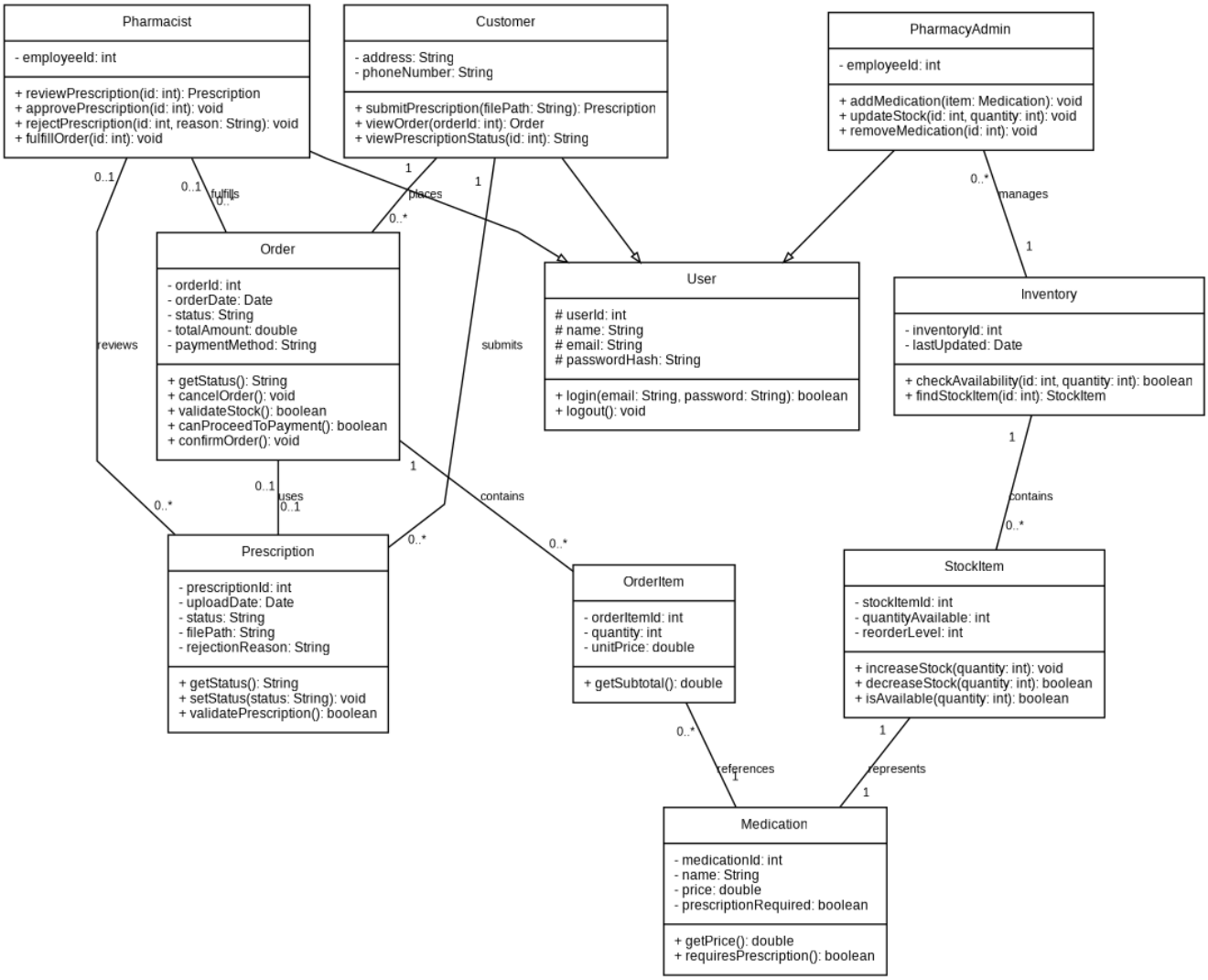
Consolidation Record and Design Decisions

Repeated class	Models involved	Resolution
User	Prescription, ordering, inventory	Use one common superclass
Customer	Prescription and ordering	Use a single customer class
Pharmacist	Prescription and ordering	Combine review and fulfillment methods
Order	Prescription, ordering and inventory	Use one order class with consistent status and checks
Medication	Ordering and inventory	Use one catalog medication class; move quantity to OrderItem and StockItem

User is the common superclass for Customer, Pharmacist and PharmacyAdmin. Prescription stores submission and review information, while Order stores purchase information. OrderItem represents the quantity requested for a particular medication in an order. StockItem tracks quantity available in inventory. The team should retain only genuine inheritance relationships and include the named associations and multiplicities shown in the combined model.

Prescription assignment to a pharmacist may be optional before a review occurs. Similarly, an order may temporarily contain no finalized order items while being assembled. These optional relationships should be checked against the team's agreed rules before final submission.

Final Consolidated UML Class Diagram



Relationship Explanation

Relationship	Meaning
Customer and Prescription	A customer may submit multiple prescriptions; each prescription belongs to one customer.
Pharmacist and Prescription	A pharmacist may review many prescriptions, but a pending prescription may not yet have a reviewer.
Customer and Order	A customer may place multiple orders; each order belongs to one customer.
Order and Prescription	An order can be associated with a prescription when prescription validation is required.
Order and OrderItem	An order groups its requested medication items.
OrderItem and Medication	Each order item identifies exactly one medication.
Inventory and StockItem	The inventory maintains zero or more stock item records.
StockItem and Medication	Each stock item refers to a medication.
PharmacyAdmin and Inventory	An administrator manages stock records within the inventory.
Pharmacist and Order	A pharmacist may fulfill orders; assignment may be pending.

Class Model Consistency and Traceability

Functionality	Use cases	Classes	Model check
Registration and login	UC21, UC22	User, Customer	Common login information is modeled in User
Prescription submission and status	UC01, UC06	Customer, Prescription	Submitted prescriptions are associated with customers
Prescription review	UC11, UC12, UC24	Pharmacist, Prescription, Order	Review status affects order eligibility
Browsing medication	UC16, UC17	Medication	Medication details are stored in Medication
Order management	UC03, UC13, UC14, UC15	Customer, Order, OrderItem	Orders store the items and quantities requested
Stock administration	UC04, UC05, UC08	PharmacyAdmin, Inventory, StockItem, Medication	Available stock is separated from medication details

Consistency with DFDs and data stores: Prescription, Order, Medication and Inventory are the main persistent domain concepts. The correspondence with the actual data stores in previous lab diagrams remains to be validated against those diagrams. The supplied report does not contain the earlier DFDs, so no verification of exact DFD names or data flows is asserted here.

Individual Contributions

Member	Model contribution
Omar Abdalla	Prescription review and approval or rejection
Alan Dsouza	Nonprescription ordering
Omar Alkhatib	Inventory and stock management
Qusai Al Tah	Proposed order and prescription coordination model; requires team confirmation

Repository Organization

The files for this lab belong in Project/Object_Modeling in the existing project repository. The relevant sections are Individual_Models, Class_Inventory.md, Integration_Notes.md, Class_Diagram and Class_Model_Consistency_Check.md. GitHub upload status and individual commit history have not been independently verified.

AI Tool Declaration

AI assistance was used to help prepare and revise the wording and proposed UML models in this document. The student team remains responsible for reviewing the technical content, confirming individual contributions and ensuring consistency with previous project deliverables.